Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/09/2025
PI Name:	Bowles, Dawn Ph.D.		
Project Title:	Radiation Resistance Conferred by Small Proline Rich Repeat Proteins (SPRRs)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Durham	**State:**	NC
Zip Code:	27710-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	04/14/2021	**End Date:**	10/13/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/13/22 per NSSC information (Ed., 6/28/22). NOTE: Period of performance changed to correct start/end dates to 4/14/2021 - 4/13/2021 per L. Barnes-Moten/HRP and NSSC information (original information showed 1/01/2021 - 12/31/2021)--Ed. 10/29/21.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0901		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	From an ongoing NASA funded study entitled “Proteomic signatures of space radiation induced cardiovascular degeneration” we have identified that cornification and keratinization pathways are significantly perturbed in hearts from galactic cosmic ray (GCR) irradiated mice. A family of small proline rich repeat proteins (SPRR) are central players in these pathways and are also essential players in reactive oxygen species (ROS) detoxification. These SPRRs are upregulated in various forms of tissue injury, including myocardial infarction and burn injury. As ROS are critical mediators of radiation induced molecular and cellular injury, we hypothesize that the upregulation of SPRR could function as a defense mechanism to mitigate the detrimental effects of radiation on the cardiovascular system. To date, this class of proteins has not been implicated in radiation injury. These proteins may be novel cell protective proteins that may be exploited as countermeasures for GCR and other types of radiation. Using strategies in our ongoing Translational Research Institute for Space Health (TRISH) funded grant entitled: “Gene Therapy Countermeasures for Detrimental Effects of Space Radiation” we will molecularly explore the overexpression of several SPRR proteins using viral vector mediated approaches in the following two aims. Aim 1: Determine the effects of altered SPRR levels on cellular survival and biological functions in a variety of radiation exposure settings. Viral vectors over-expressing or down-regulating SPRRs will be developed and utilized in a variety of cell models to assess the impact of individual modulation on a variety of biological outcomes post-radiation treatment. This aim will test our hypothesis that modulation such as over-expression of any of these proteins will provide resistance to radiation exposure and will provide a quick means of down-selection of the targets. Radiation exposure experiments will occur at our institution. Aim 2: Determine the effects of modulation of SPRR levels on animal survival, or physiology following exposure to GCR at Brookhaven National Laboratory (BNL). Using viral vectors based on the SPRR proteins down-selected in Aim 1 we will test the hypothesis that chosen viral vectors when administered systemically, or to targeted organs such as the heart or brain, will confer overall or organ specific resilience to radiation exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This technology has the potential to serve as a radioprotective agent for individuals undergoing cancer radiation therapy or in occupational exposure scenarios. We filed an invention disclosure on this technology to Duke OLV.
Task Progress:	From a previously NASA-funded study entitled, “Proteomic signatures of space radiation induced cardiovascular degeneration”, we have identified that cornification and keratinization pathways are significantly perturbed in hearts from Galactic Cosmic Ray (GCR) irradiated mice. A family of small proline rich repeat proteins (SPRR) are central players in these pathways, and are essential players in reactive oxygen species (ROS) detoxification. These SPRRs are upregulated in various forms of tissue injury, including myocardial infarction and burn injury. As ROS are critical mediators of radiation induced molecular and cellular injury, we hypothesize that the upregulation of SPRR could function as a defense mechanism to mitigate the detrimental effects of radiation on the cardiovascular system. To date, this class of proteins has not been implicated in radiation injury. These proteins may be novel cell protective proteins that may be exploited as countermeasures for GCR and other types of radiation. Using a rodent tissue repository established by our lab, western blotting was performed on homogenized heart apexes for 150 cGy GCR5-ion and sham irradiated controls for 12 month time point, post radiation, to gain an understanding of SPRR levels in cardiac tissue following radiation. Overall abundance of many of the SPRRs were reduced in GCR mice compared to unirradiated control mice. To test the hypothesis that over expression of SPRRs would provide protection from radiation induced injury to cells, we developed both 293 cell lines stably expressing various SPRRs, as well as recombinant adeno-associated viral vectors (rAAVs) expressing the same SPRRs, and utilized these reagents in a variety of cell based assays. Results are awaiting publication, but the data indicate that several of these SPRRs provide protection from detrimental effects of radiation providing increased cell viability, resistance to apoptosis, and sequestration of ROS.
Bibliography Type:	Description: (Last Updated: 06/30/2026)